# KRASNER'S LEMMA FOR A COMPLETE LOCAL DOMAIN OF DIMENSION ONE: A SIMPLE ROOT SURVIVES EVERY SUFFICIENTLY SMALL PERTURBATION OF THE POLYNOMIAL

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*Editorial note.* The proof writes the inverse of P'(alpha) as pi^{-m} a and uses the macro lim, which the retained preamble supplies; nothing outside the counted spans is required to read the delivered argument.

*Allowed standard prerequisites (editorial).*

- In a local domain of dimension one the maximal ideal is the radical of a principal ideal generated by any nonzero element of it. used at proof lines 33758-33759, where a nonzero pi in the maximal ideal is fixed and m = sqrt((pi)) is recorded. Turns the dimension-one hypothesis into the valuation-like statement that an element of the fraction field with a pole can be written as pi to a negative power times an element of A, which is what produces the integer m with P'(alpha)^{-1} = pi^{-m} a.
- The Taylor expansion of a polynomial in two variables, P(x + y) = P(x) + P'(x) y + R(x, y) y^2 for some R in A[x, y]. recorded at proof lines 33763-33766 and used again at proof lines 33792-33796. Supplies the quadratic error term that makes the Newton correction work modulo the next power of the maximal ideal.
- Completeness of a local ring with respect to its maximal-adic filtration: a sequence whose consecutive differences lie in higher and higher powers of

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the maximal ideal converges, and its limit can be used in A. used at proof line 33774, where beta is defined as the limit of the constructed sequence. Converts the inductively constructed sequence into an actual element beta of A with the required congruence beta - alpha in m_A^c.

**Lemma 0.2** (Krasner's lemma)**.** *Let $A$ be a complete local domain of dimension 1. Let $P(t) \in A[t]$ be a polynomial with coefficients in $A$. Let $\alpha \in A$ be a root of $P$ but not a root of the derivative $P' = dP/dt$. For every $c \geq 0$ there exists an integer $n$ such that for any $Q \in A[t]$ whose coefficients are in $\mathfrak{m}_A^n$ the polynomial $P + Q$ has a root $\beta \in A$ with $\beta - \alpha \in \mathfrak{m}_A^c$.*

*Proof.* Choose a nonzero $\pi \in \mathfrak{m}$. Since the dimension of $A$ is 1 we have $\mathfrak{m} = \sqrt{(\pi)}$. By assumption we may write $P'(\alpha)^{-1} = \pi^{-m}a$ for some $m \geq 0$ and $a \in A$. We may and do assume that $c \geq m + 1$. Pick $n$ such that $\mathfrak{m}_A^n \subset (\pi^{c+m})$. Pick any $Q$ as in the statement. For later use we observe that we can write

$$P(x + y) = P(x) + P'(x)y + R(x, y)y^2$$

for some $R(x, y) \in A[x, y]$. We will show by induction that we can find a sequence $\alpha_m, \alpha_{m+1}, \alpha_{m+2}, \ldots$ such that

(1) $\alpha_k \equiv \alpha \bmod \pi^c$,
(2) $\alpha_{k+1} - \alpha_k \in (\pi^k)$, and
(3) $(P + Q)(\alpha_k) \in (\pi^{m+k})$.

Setting $\beta = \lim \alpha_k$ will finish the proof.

Base case. Since the coefficients of $Q$ are in $(\pi^{c+m})$ we have $(P + Q)(\alpha) \in (\pi^{c+m})$. Hence $\alpha_m = \alpha$ works. This choice guarantees that $\alpha_k \equiv \alpha \bmod \pi^c$ for all $k \geq m$.

Induction step. Given $\alpha_k$ we write $\alpha_{k+1} = \alpha_k + \delta$ for some $\delta \in (\pi^k)$. Then we have

$$(P + Q)(\alpha_{k+1}) = P(\alpha_k + \delta) + Q(\alpha_k + \delta)$$

Because the coefficients of $Q$ are in $(\pi^{c+m})$ we see that $Q(\alpha_k + \delta) \equiv Q(\alpha_k) \bmod \pi^{c+m+k}$. On the other hand we have

$$P(\alpha_k + \delta) = P(\alpha_k) + P'(\alpha_k)\delta + R(\alpha_k, \delta)\delta^2$$

Note that $P'(\alpha_k) \equiv P'(\alpha) \bmod (\pi^{m+1})$ as $\alpha_k \equiv \alpha \bmod \pi^{m+1}$. Hence we obtain

$$P(\alpha_k + \delta) \equiv P(\alpha_k) + P'(\alpha)\delta \bmod \pi^{k+m+1}$$

Recombining the two terms we see that

$$(P + Q)(\alpha_{k+1}) \equiv (P + Q)(\alpha_k) + P'(\alpha)\delta \bmod \pi^{k+m+1}$$

Thus a solution is to take $\delta = -P'(\alpha)^{-1}(P + Q)(\alpha_k) = -\pi^{-m}a(P + Q)(\alpha_k)$ which is contained in $(\pi^k)$ by induction assumption. □